Participant Protection in Stable Schedule Repair

line 1: 1st Given Name Surname
line 2: *dept. name of organization (of Affiliation)*
line 3: *name of organization (of Affiliation)*
line 4: City, Country
line 5: email address or ORCID

line 1: 2nd Given Name Surname
line 2: *dept. name of organization (of Affiliation)*
line 3: *name of organization (of Affiliation)*
line 4: City, Country
line 5: email address or ORCID

line 1: 3rd Given Name Surname
line 2: *dept. name of organization (of Affiliation)*
line 3: *name of organization (of Affiliation)*
line 4: City, Country
line 5: email address or ORCID

line 1: 4th Given Name Surname
line 2: *dept. name of organization (of Affiliation)*
line 3: *name of organization (of Affiliation)*
line 4: City, Country
line 5: email address or ORCID

line 1: 5th Given Name Surname
line 2: *dept. name of organization (of Affiliation)*
line 3: *name of organization (of Affiliation)*
line 4: City, Country
line 5: email address or ORCID

line 1: 6th Given Name Surname
line 2: *dept. name of organization (of Affiliation)*
line 3: *name of organization (of Affiliation)*
line 4: City, Country
line 5: email address or ORCID

Abstract—After a workforce disruption, equally stable schedule repairs can affect different people. We present a coordinator–participant protocol that distinguishes incidental satisfaction, checked protection commitments, solver-confirmed refusals and unresolved search, while tracking adopted repairs and offering bounded concessions. On 50 public project models over 20 simulated workdays, Exact and twenty-sample coordination both deliver 88.7%; Exact confirms 113 infeasible requests left unresolved by sampling. At matched daily stability, Random, Round-robin, Least-recent and Burden-aware already satisfy about 93–96% of acceptable requests without protection constraints. With requests enforced, Burden-aware lowers the mean of each model's maximum personal impact count from Random's 4.28 to 3.30 days. Its advantage over Least-recent varies across sensitivity checks. These results support combining explicit request handling with history-aware selection.

Keywords—collaborative scheduling, participant requests, schedule repair, cumulative impact

I. Introduction

After an outage, two coworkers ask to retain their duties, but only one can be protected. What can the coordinator promise, what concession could honor the other request, and who absorbs the next disruption? Repairs affecting {a,b,c} and {a,b,d} can be equally stable, yet distribute impact differently.

In Figure 1, W8 and W11 each have a protecting repair at the same optimum; W2 does not. The protocol supports participants submitting requests and the coordinator reviewing response states, authorizing concessions, adopting a checked repair and retaining its impact history.

We evaluate protection requests at proved repair optima and track the cumulative impact of adopted responses. We contribute:

(1) A coordinator–participant protocol distinguishing checked protection, solver-confirmed refusal and UNKNOWN, with bounded, witnessed concessions.

(2) A characterization of individual protection and joint compatibility within the stability-optimal repair space.

(3) A matched request × selection evaluation separating explicit commitments from incidental delivery and repeated-impact allocation.

II. Related Work

Collaborative scheduling. CSCWD covers urgent workforce allocation [1] and disruption recovery [2]. Recent studies examine driver rostering [3], fairness-aware team formation [4] and multi-weight fair assignment [5]. Our protocol retains participants' duties within a proved repair boundary and carries impact across repeated decisions. Self-scheduling [6, 7], interactive conflict resolution [8] and decision support [9] motivate its coordinator–participant workflow.

Requests and explanations. EXPRES [10] explains unsatisfied scheduling preferences and includes employee evaluation. Industrial workforce scheduling [11] combines load balancing, MUS/MCS and recovery. CMAOE [12] generates alternatives under requested properties, assuming the augmented problem is solvable; joint requests are expressible there. Our evaluation fixes proved repair-stability optima and separately measures delivered protection, solver-confirmed refusal, UNKNOWN and checked budget concessions. Constraint explanation [13, 14] and scheduling diagnosis [15] provide the reasoning foundations.

Residual choice and history. Backbone analysis [16], diverse solutions [17] and minimal perturbation [18] underlie protection maps. Rostering trade-offs [19] and procedural justice [20] distinguish computational burden measures from perceived fairness. Historical scheduling [21, 22] motivates incorporating past outcomes; we evaluate that choice within the daily repair boundary across repeated workdays.

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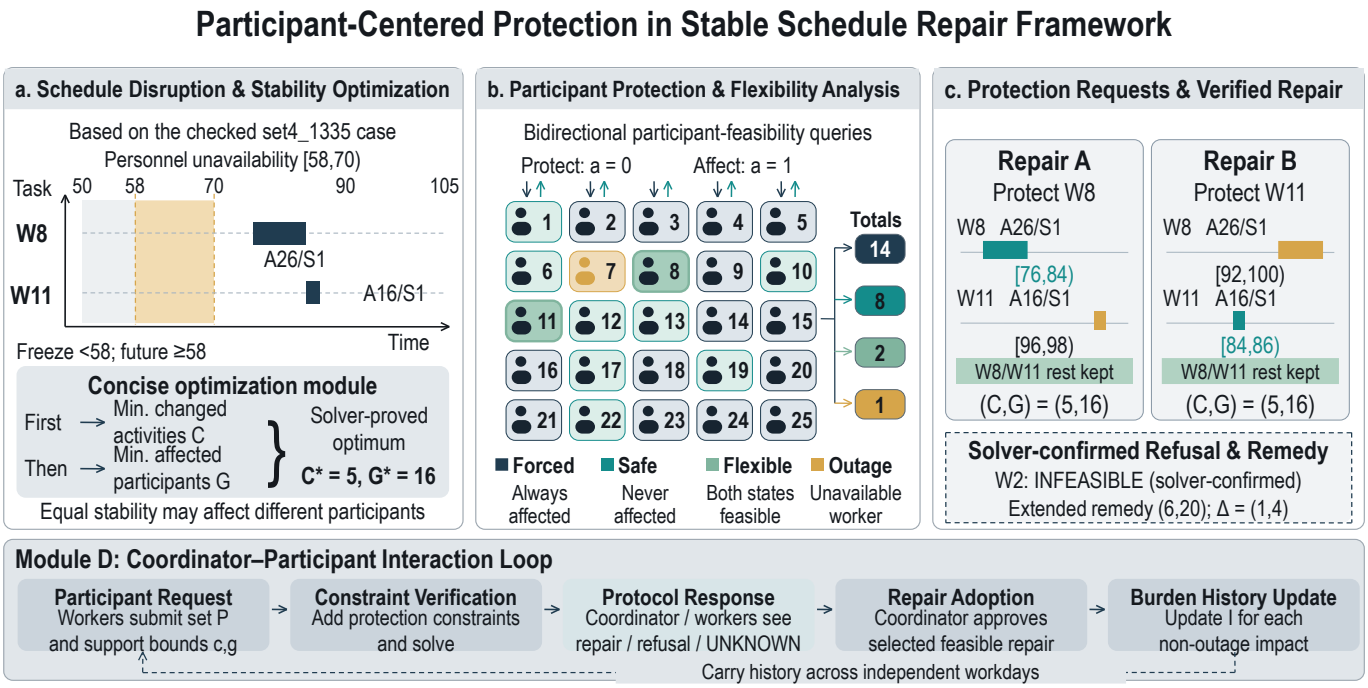


Fig. 1. Coordinator–participant protocol and checked set4_1335 example. The 5×5 grid contains workers 1–25 (14 forced, eight safe, two flexible, one unavailable). Repairs A/B preserve W8/W11 respectively at (C,G) = (5,16); the other duties of W8/W11 stay unchanged. W2's extended remedy, ΔC = 1 and ΔG = 4, lies outside the concession menu used in sessions.

III. Repair Model and Protection

The reference schedule x^0 has makespan T_0 and assigns nonpreemptive activities to skilled workers with integer starts, precedence and no overlapping duties. An outage makes worker o unavailable over an interval. Starts and crews of activities beginning before disruption time t_d are frozen; remaining activities start no earlier than t_d. Repairs must finish by deadline H. These execution constraints remain hard in operational queries.

TABLE I. Core Notation

Symbol	Meaning
C, G	Changed activities; affected participants
a_w	Exact participant impact indicator
C^*, G^*; F*	Proved lexicographic optimum; its repairs
P; c,g	Protection request; change-budget caps
I_w	Prior non-outage impact days
F; k	Flexible participants; number impacted

C counts activities whose start or crew changes. $a_w = 1$ exactly when w gains/loses a duty, changes skill role, or has a retained duty's start changed; coworker changes alone do not affect w. $G = \sum a_w$. First prove minimum C^*, then minimum G^* at C^*. F* contains all repairs with these values.

At C^*,G^*, queries $a_w = 0$ and 1 classify non-outage workers as forced (protection infeasible), safe (impact infeasible) or flexible (both feasible). UNKNOWN leaves them unclassified. A map uses at most $2(|W|-1)$ calls. With complete classification, affected sets vary iff F is nonempty.

Proposition 1. Fix proved C^*,G^*, complete non-outage classification and outage-impact flags. Let O contain impacted outage workers. Exactly $k = G^* - n_f - |O|$ flexible workers are impacted, where n_f counts forced non-outage workers. If $k = 1$, every proper subset of F is protectable. Proof: forced/outage impact is $n_f + |O|$; safe workers contribute zero. Each flexible worker can be the sole flexible impact; choose one outside P to protect P. For $k > 1$, capacity alone is insufficient.

IV. Coordinator–Participant Protocol

A. Respond, offer remedies and adopt

Participants submit a set P and caps c,g. A protection query asks for a repair with $a_w = 0$ for all w in P and $C \le c$, $G \le g$. A feasible response includes a schedule checked independently of its encoding. INFEASIBLE yields a solver-confirmed refusal; a timed-out query is UNKNOWN. The coordinator and requesters receive this response; the coordinator authorizes concessions and adopts a checked repair.

Let $\mathcal{F}(c,g)$ contain all repairs satisfying the hard constraints and $C(x) \le c$, $G(x) \le g$, without protection constraints. Residual requested impact is

$$R(c,g;P) = \min_{x \in \mathcal{F}(c,g)} \sum_{w \in P} a_w(x)$$

Set $R = +\infty$ for empty $\mathcal{F}$; otherwise $R = 0$ iff P is jointly protectable. R is nonincreasing in each cap. Search $\Delta C, \Delta G \le \Delta = 4$ with $O(\Delta)$ staircase calls in resolved boxes. Confirm frontier points only after ruling out cheaper dominating budgets; UNKNOWN leaves the frontier incomplete.

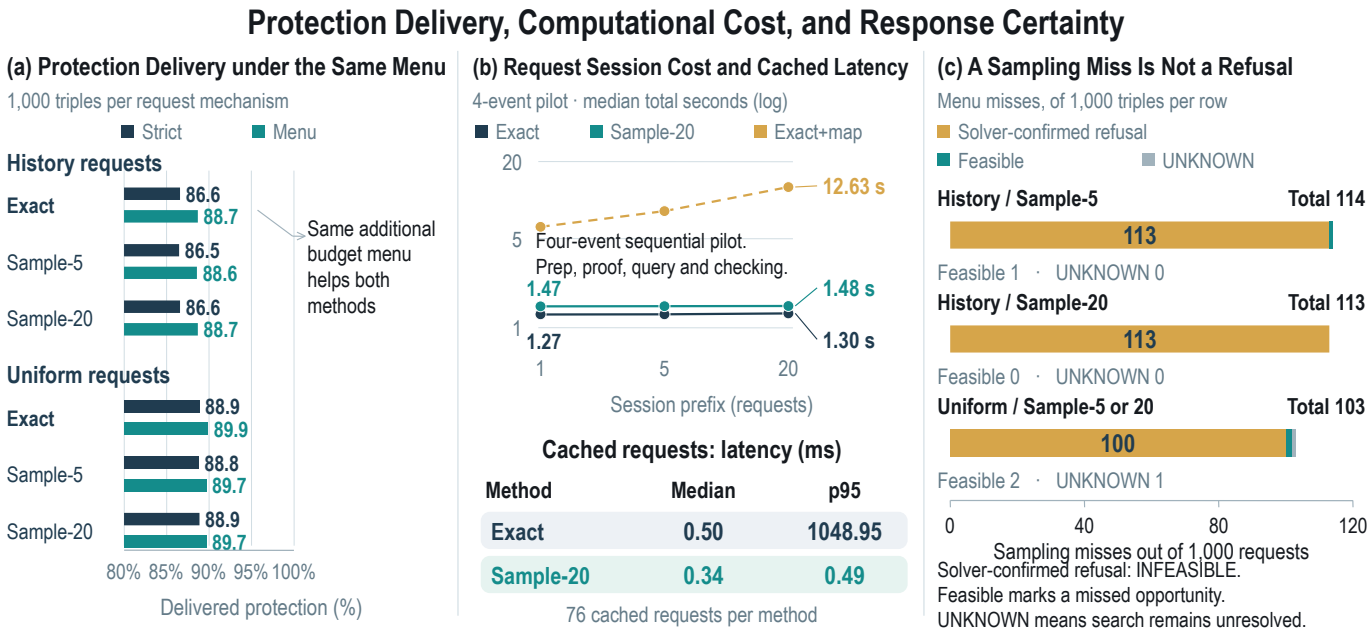


Fig. 2. Same-menu protection delivery and response certainty for 1,000 triples per mechanism. (a) Strict and menu delivery; the percentage axis starts at 80%. (b) Response-session cost in a four-event sequential pilot, with direct cached-latency summaries (76 subsequent requests/method); daily history selection is excluded. (c) Exact decomposes total sampling misses; solver-confirmed refusals are distinct from independently checked arithmetic certificates.

Sessions use the concession menu (0,0), (0,1), (1,0), (1,1), relative to the proved optimum. The first checked protecting repair is accepted. If all entries are infeasible, return a refusal and retain the checked strict repair; unresolved searches remain UNKNOWN. After adoption, increment I_w for every affected non-outage worker, including extra impact from concessions.

Core diagnostics guard requests, freezes and C/G caps against hard constraints; unselected guards are disabled. Remove guards only after INFEASIBLE. Subset-minimality requires feasibility after every remaining deletion; UNKNOWN leaves minimality unresolved.

B. Choose who absorbs the change

At the chosen daily C,G, First (the default solver choice) takes the first feasible repair. Random minimizes seeded impact weights 1–10. Burden-aware gives less priority to disrupting people already affected often: over non-outage workers it minimizes $\sum(1+I_w)a_w$. At fixed G and outage flags, this minimizes the one-step increase in $\sum I_w^2$, since $(I_w+a_w)^2-I_w^2 = (2I_w+1)a_w$. Each policy maintains its own adopted impact history.

Round-robin minimizes cyclic ranks, advancing past the furthest affected rank: choosing W2,W3 in ring W1…W4 makes W4 next (outage worker skipped). Least-recent minimizes last-impact dates with worker-ID ties. All policies share a twenty-second construction/search limit. Saved optima are reused only at identical constraints and weights; other calls solve anew, retaining a checked common witness on failure.

V. Evaluation Design

We use MSLIB [23, 24], excluding twenty development models and deduplicating normalized inputs before seeded size-stratified sampling (25/25/10 models at 30/60/90 activities). The package records the asset SHA-256 and filenames; v2.2 upstream prose still refers to v2.1. CP-SAT [25] 9.11 uses one worker, seed 41 and an i7-14650HX. Base proof limits are 4/30/60 seconds; workday setup uses 30/60 seconds and selection twenty. Independent instances run in separate processes.

The fifty 30/60-activity models supply independent workdays, retaining only burden history. Ninety-activity cases assess scaling. $H = \lceil 1.5T_0 \rceil$; outage duration is $\max(1, \text{round}(0.08T_0))$. A frozen candidate order selects proved events, replacing twelve unproved candidates. History requests target First's three most impacted non-outage workers; Uniform uses independent seeded permutations. Table II fixes History-selected actual C,G; Figure 2 evaluates each mechanism's menu. Own-history replay uses each policy's adopted history.

Exact and samplers share requests, menu order and caps. Request-independent pools of K = 5 or 20 repairs are reusable within an event/cap; misses remain unresolved. The four-event sequential cost pilot includes preparation, proofs, pools, queries and checking for 1/5/20 requests.

We compute 95% confidence intervals using 10,000 size-stratified model bootstrap resamples. The primary measure is each model's largest personal impact count. SD, Gini and distinct workers assess dispersion; cumulative shift sums absolute start changes per retained worker–activity pair. Actual G controls total impact.

Own-history replay adds Least-recent and Round-robin to archived First/Random/Burden-aware on all fifty models. Each policy requests its own top-three; queries and selection each receive four seconds. Requests and budgets diverge; outcomes, costs, impact and status are reported separately.

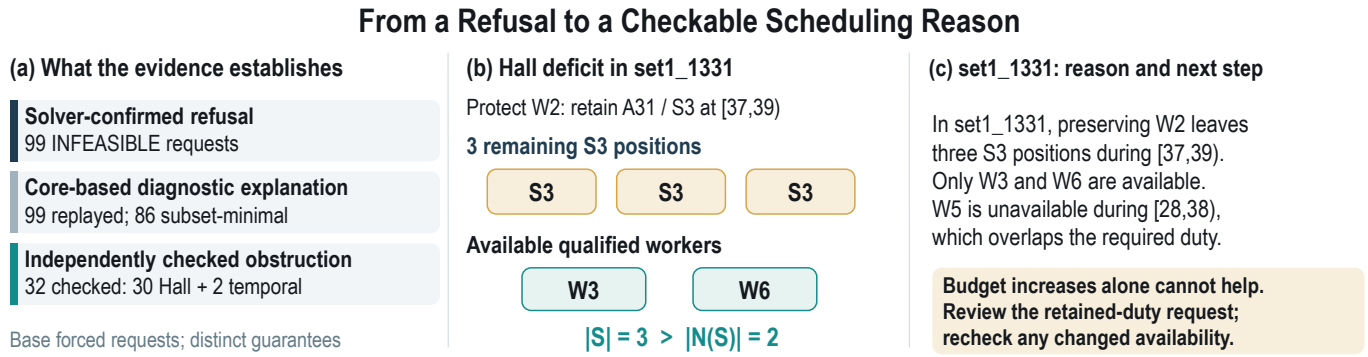


Fig. 3. Three evidence levels in set1_1331 (distinct from Fig. 1's set4_1335): solver-confirmed refusal, core-based diagnostic explanation and independently checked obstruction. Panel (c) is an illustrative response template. Preserving W2's A31/S3 duty at [37,39) leaves three S3 positions but only two eligible workers while W5 is unavailable during [28,38). All 32 certificate requests remain solver-INFEASIBLE without C/G caps; the Hall arithmetic is independently checked.

VI. Results

A. Requests receive decisions, not just candidate lists

Exact and Sample-20 deliver 86.6% strictly and 88.7% under the concession menu (Fig. 2a), with 21 rescued requests. At equal delivery, Exact confirms all 113 remaining History requests infeasible; Sample-20 leaves them unresolved. Sample-5 misses 114, including one feasible request. Uniform menu delivery is 89.9%/89.7% for Exact/Sample-20; its 103 sampling misses comprise 100 refusals, two feasible requests and one unresolved query. A checked sampled repair also supports a positive protection commitment.

On four cost-pilot events, median total time is 1.27/1.47 s for one Exact/Sample-20 request and 1.30/1.48 s for twenty requests. A complete Exact map raises the latter median to 12.63 s. Cached median/p95 latency is 0.50/1,048.95 ms for Exact and 0.34/0.49 ms for Sample-20. The four-event pilot provides a local comparison of session and cached-query costs.

B. Protection constraints and selection have separate effects

TABLE II. Matched request × selection at identical daily C,G (20 s).

(a) Maximum personal impact count (days)

Protection	First	Random	Round-robin	Least-recent	Burden-aware
No request	7.96	4.34	3.70	3.46	3.36
Enforced	6.56	4.28	3.80	3.54	3.30

(b) Delivered without protection constraints

Request cohort	First	Random	Round-robin	Least-recent	Burden-aware
History /1,000	479	838	826	846	837
Uniform /1,000	844	849	860	857	852
Flexible /278	74	240	232	243	237

History/Uniform: 887/893 acceptable at History-selected C,G (1,000 requests each). Uniform's own menu delivers 899 (Fig. 2a). Flexible: 278 acceptable History requests with a verified flexible member.

At common budgets, First naturally delivers 54.0%/94.5% of acceptable History/Uniform requests; the other rules deliver 93.1–95.4%/95.1–96.3%. Enforcing protection adds 408 deliveries for First and 41–61 for each alternative rule (50 for Burden-aware). History requests target First's prior impacts, so its gain is not representative. Burden-aware's maximum-impact change is −0.06 days (95% CI [−0.32,0.12]); Round-robin/Least-recent change by +0.10/+0.08.

The 278 acceptable History requests with witnessed flexible members have mean f = 10.17; 273 are on the strict face (mean k = 1.04). For each saved pool let f count witnessed flexible workers, h their impacts per repair and m their request overlap. The idealized uniform-subset reference is comb(f−m,h)/comb(f,h), averaged equally over these 278 requests: 82.7%. Random delivers 240/278 (86.3%) on the same requests. Random-weight optimization is not uniform subset sampling; the reference omits unwitnessed members and compatibility constraints. The other 609 acceptable requests remain unclassified, not certified safe.

C. Protection structure and evidence for refusals

At 30/60/90 activities, C,G optima are proved for 25/25, 25/25 and 9/10 models; median recorded map-construction times are 9.57/27.09/92.21 seconds (summed classification calls; baseline proof/checks excluded; mixed limits). The 59 maps classify 99 forced, 1,658 flexible and 3,964 safe participants. All 44 variable base problems have k = 1. Fresh single-outage days have k = 0 on 203/1,000 events and k ≥ 2 on fourteen. Of 134 strict history-triple refusals, 129 contain a forced singleton and five are joint conflicts. The concession menu leaves 113 refusals, including fifteen joint conflicts.

Evidence has three levels (Fig. 3): solver-confirmed refusal, core-based diagnosis against hard constraints, and independently checked Hall [26] or temporal obstruction. All 99 cores replay, 86 are subset-minimal and 63 contain only the request. Arithmetic obstructions cover 32/99 (30 Hall, two temporal); the other 67 retain solver results and core diagnostics.

In set1_1331, reserving W2 leaves three S3 positions but only W3 and W6 eligible during [37,39) (Fig. 3). Removing C/G caps rescues 22/99 requests, adding unfreezing rescues 35, and extending the deadline alone rescues two. All 32 certificate requests remain infeasible without C/G caps. Categories overlap; unfreezing is diagnostic.

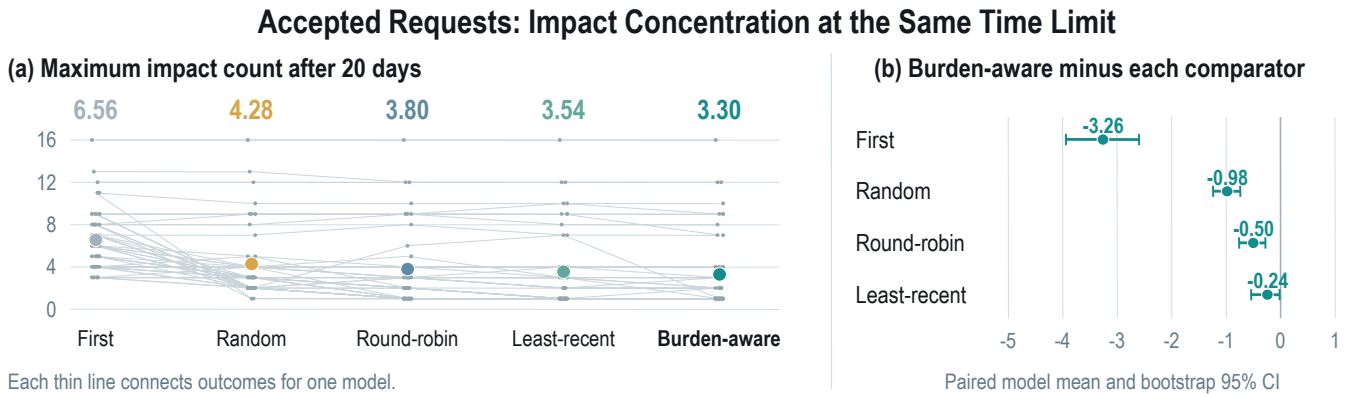


Fig. 4. Accepted requests at common actual daily C,G and a twenty-second selection limit. All policies deliver 887/1,000. Thin lines connect outcomes for one model; large points are means. Paired differences show Burden-aware minus each comparator, with model-bootstrap 95% intervals.

D. *The same delivered requests allow less concentrated impact*

Table III and Fig. 4 evaluate accepted requests at identical actual daily C,G: totals C = 2,235 and G = 2,794 for every policy. All deliver the same 887 requests, with 113 common refusals and zero unresolved responses. Burden-aware reduces maximum personal impact from Random's 4.28 to 3.30; the paired difference is -0.98 days (95% CI [-1.24,-0.74]), with 35/14/1 better/tied/worse models. SD and Gini assess dispersion at the same total impact.

TABLE III. ACCEPTED REQUESTS, 20-SECOND LIMIT: MODEL MEANS. SHIFT = MAXIMUM PERSONAL CUMULATIVE START-TIME CHANGE (BENCHMARK TIME UNITS); I>0 = WORKERS EVER AFFECTED. O/F/B = OPTIMAL / FEASIBLE UNPROVED / CHECKED FALLBACK (1,000 DAYS EACH). FIRST'S F IS FEASIBILITY ONLY; MATCHING SAVED OPTIMA COUNT IN O.

Policy	Max I	SD	Gini	I>0	Shift	O/F/B
First	6.56	1.24	0.765	11.66	84.78	—/981/19
Random	4.28	0.99	0.705	17.32	81.88	997/1/2
Round-robin	3.80	0.96	0.687	18.22	83.18	982/0/18
Least-recent	3.54	0.90	0.652	19.64	84.00	980/0/20
Burden-aware	3.30	0.88	0.649	19.92	83.12	999/0/1

Burden-aware versus Round-robin is −0.50 days (95% CI [−0.76,−0.28]; 18/32/0 better/tied/worse models); Least-recent versus Round-robin is −0.26 ([−0.44,−0.10]). Burden-aware versus Least-recent is −0.24 ([−0.54,−0.02]). Fallbacks are 18/20/1 for Round-robin/Least-recent/Burden-aware. Recounting the original trajectories only on 977 jointly OPTIMAL days, including Random, those three policies yield 3.68/3.38/3.30. Burden-aware differs from Round-robin by −0.38 ([−0.54,−0.24]) and from Least-recent by −0.08 ([−0.20,0.04]). This conditional sensitivity retains the original decision histories, including earlier fallbacks; it is not a rerun without solver effects. With identical saved candidate pools and no selector solver calls, Least-recent/Burden-aware yield 3.40/3.30; their paired gap is −0.10 days (95% CI [−0.22,0.00]). The common-pool Round-robin mean is 4.04 (Burden-aware 3.30).

E. *Closed-loop sensitivity and supporting checks*

Own-history outcomes are shown in Table IV. Burden-aware and Least-recent tie on maximum count in all fifty models, but deliver 877/880 requests at total C = 2,229/2,233 and G = 2,792 each. Recorded response/selection costs and O/F/B counts are in the versioned artifact. Supporting no-request tests give Minimax/Burden-aware 3.34/3.32 (49/50 ties); Count-first (2+2 seconds) gives count/shift 3.38/60.52 versus 3.32/81.84. Offline column upper bounds and global relaxation lower bounds prove the archived four-second Burden-aware final maximum count optimal in 20/25 small models; mean excess lies in [0.16,0.24].

TABLE IV. OWN-HISTORY REPLAY (4 S/QUERY AND SELECTION). D/R/U = DELIVERED/REFUSED/UNRESOLVED; MENU = DAYS ADOPTING CONCESSIONS. REQUESTS AND ACTUAL C,G VARY BY POLICY; SHIFT USES BENCHMARK TIME UNITS.

Policy	D/R/U	Max I	Shift	ΣC/ΣG	Menu
First	877/123/0	4.76	82.62	2231/2796	19
Random	877/121/2	3.74	84.10	2231/2793	18
Round-robin	881/119/0	3.30	83.62	2233/2793	19
Least-recent	880/120/0	3.18	84.08	2233/2792	19
Burden-aware	877/123/0	3.18	82.36	2229/2792	14

The Poisson stress test has 95/100 proved events, 85 complete maps and 35/85 with k ≥ 2. It rejects 54/105 at-capacity groups and all 105 above-capacity groups. Twenty preselected two-outage cases include fourteen with k = 2; of 40,542 affected pairs, 19,830 are compatible, 20,701 incompatible and eleven UNKNOWN. Queries protect F minus the pair. The constructed cohorts test protection near and above the compatibility boundary.

VII. DISCUSSION AND FUTURE WORK

Collaborative shift scheduling reconciles coworkers' requests [7, 8], while procedural justice examines transparency and outcome control [20]. The protocol supports explicit responses and coordinator authorization. Equal impact counts do not measure subjective harm. Human acceptance and perceived fairness were not evaluated.

History requests target First's prior impacts, so its delivery gain is specific to this stress test. Deployment requires request priorities and appeals; strategic request behaviour is outside this evaluation. Constructed outage cohorts do not estimate field frequencies.

The protocol separates a checked protection commitment, a solver-confirmed refusal and an unresolved search. In matched simulations, selection satisfies many requests incidentally; protection constraints secure the remaining feasible requests. Burden-aware reduces repeated impact relative to Random and Round-robin, while its advantage over Least-recent is not robust across all checks. These findings support combining explicit request handling with history-aware choice.

Future user evaluations should compare refusal-only responses, diagnostic reasons and checked concessions in real coordination workflows [27].

Computational artifacts: https://github.com/sxxl0326-collab/participant-protection-schedule-repair/tree/main/versions/v35. Evidence identifier: PP-SR-20261101-v35.

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